

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Duration :60 Mins

Total Marks:50

Assessment Mode: In Class

Annexure A

CONCEPT MAPPING

Code of Conduct:

I SHEIK MOHAMED M 192511145 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

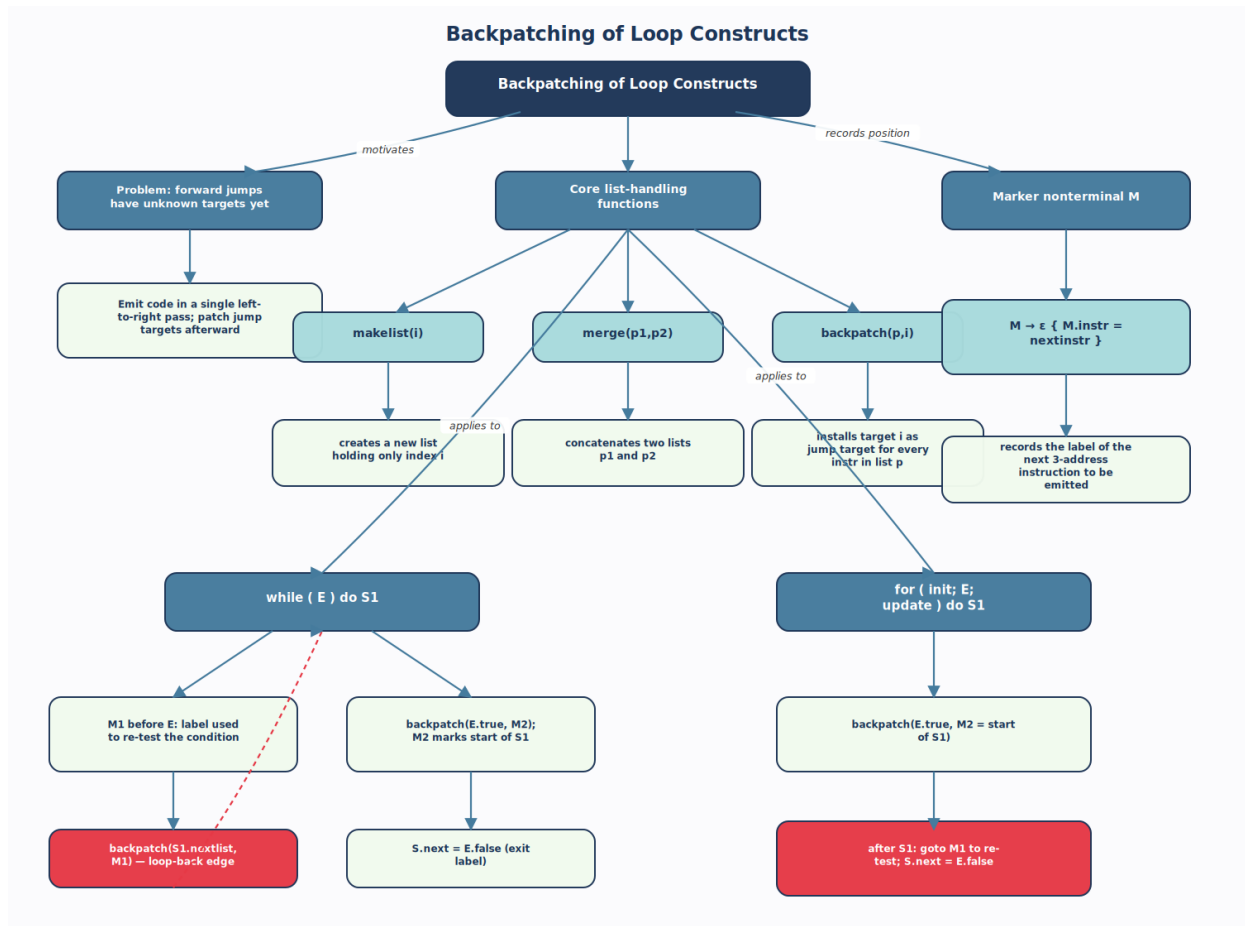
Signature of the student:

SHEIK MOHAMED M

Q. No	Understanding of Data (25)	Analysis (25)	Application of Concepts (20)	Conclusion (20)	Clarity (10)	(out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVER ALL RUBRIC'S VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 20	/ 20	/ 10	/ 100

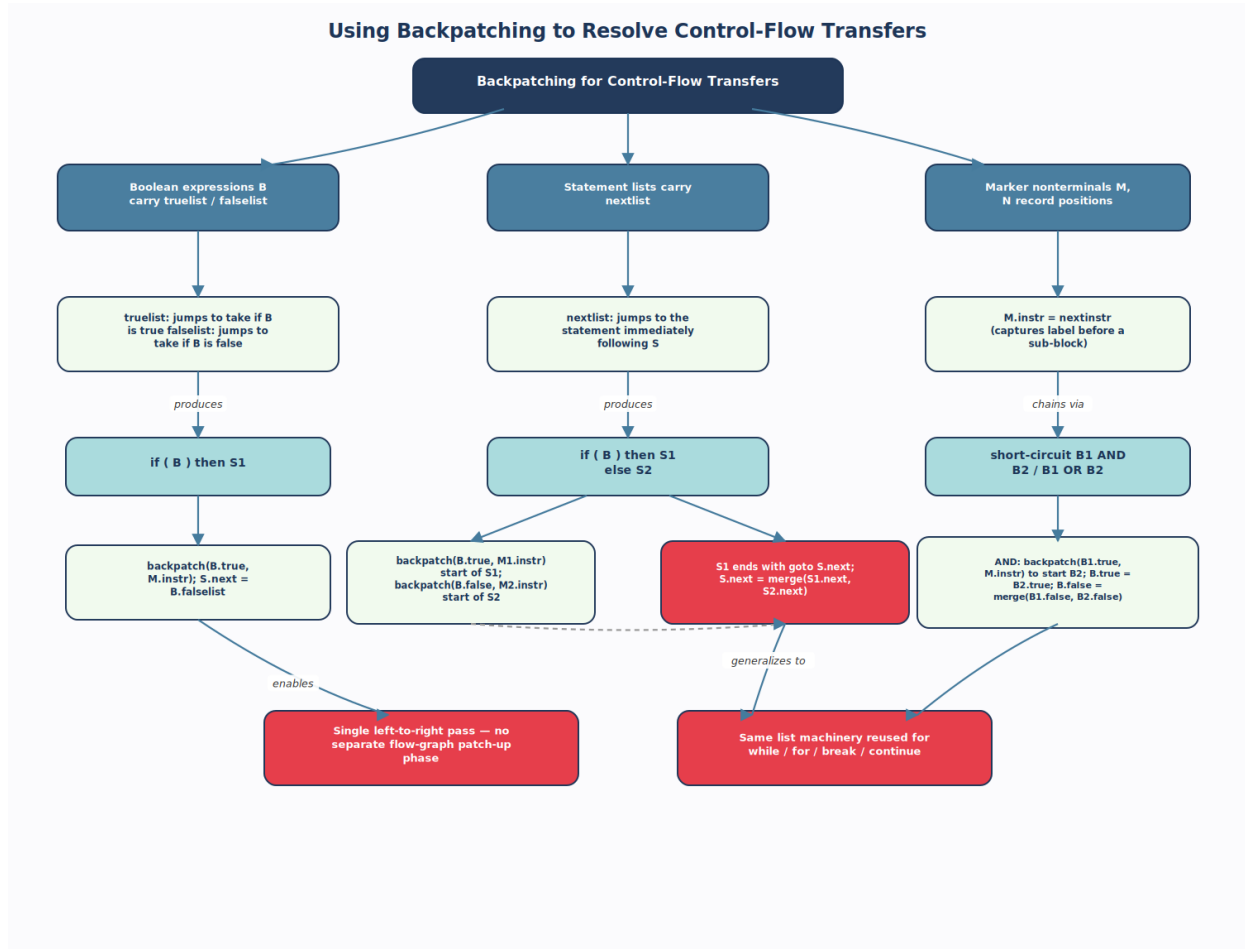
Annexure A

30. Backpatching of Loop Constructs



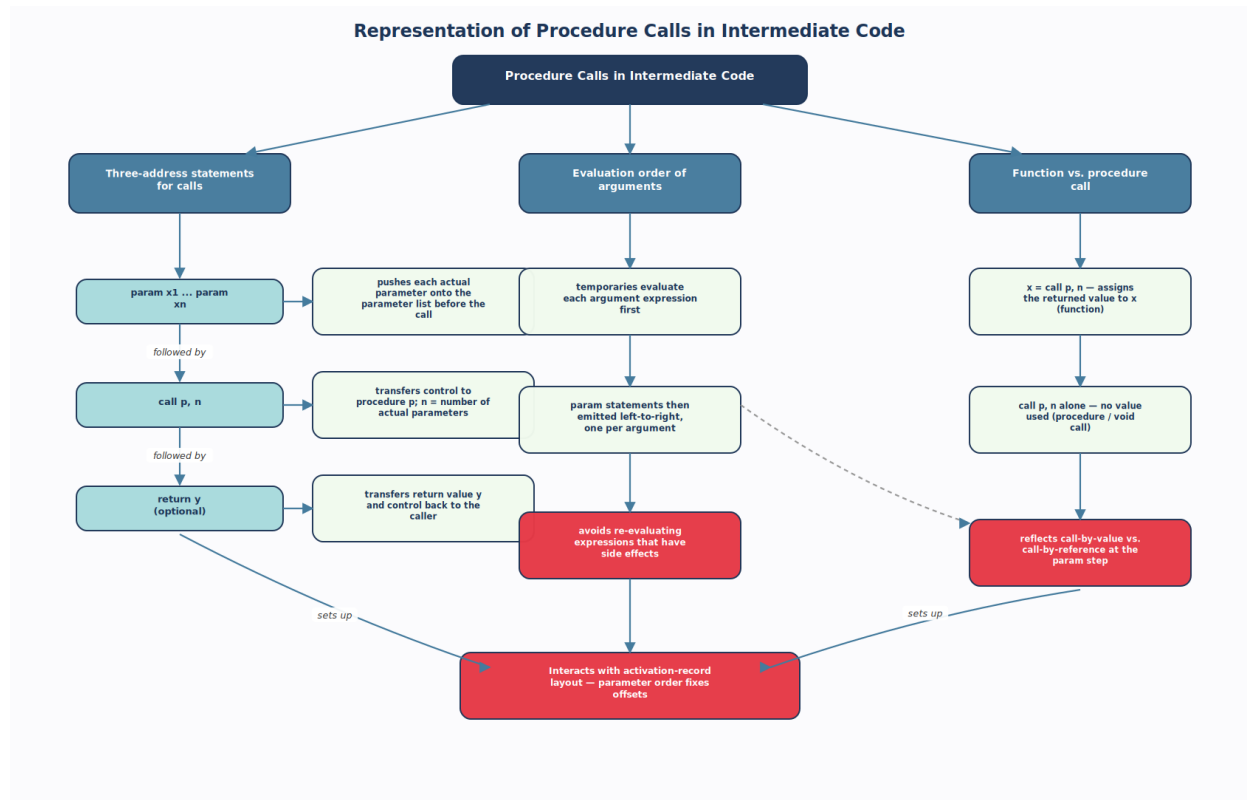
Concept map: functions makelist, merge, and backpatch are combined with marker nonterminals to fill in jump targets for while- and for-loop translation in a single left-to-right pass.

31. Using Backpatching to Resolve Control-Flow Transfers



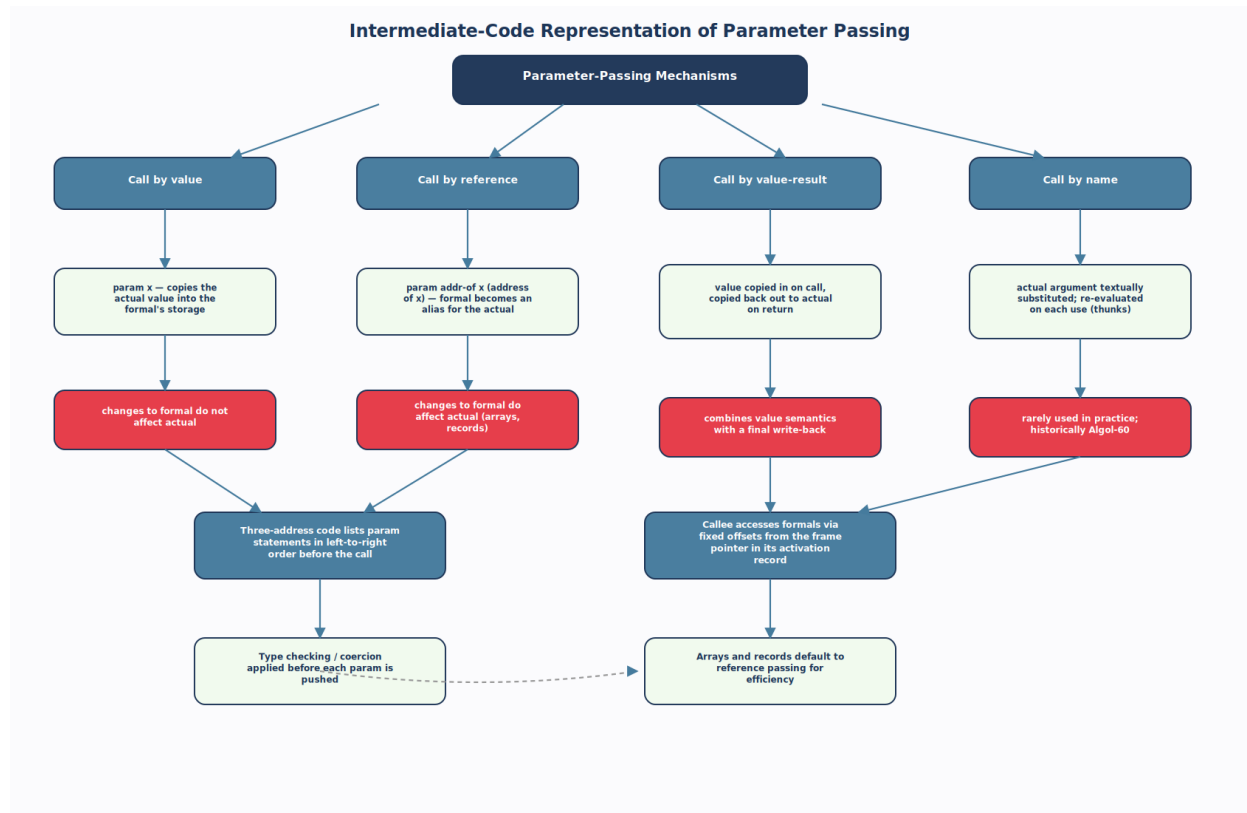
Concept map: truelist/falselist and nextlist attributes, together with marker nonterminals, let if-then, if-then-else, and short-circuit boolean expressions be translated without a separate flow-graph patch-up phase.

32. Representation of Procedure Calls in Intermediate Code



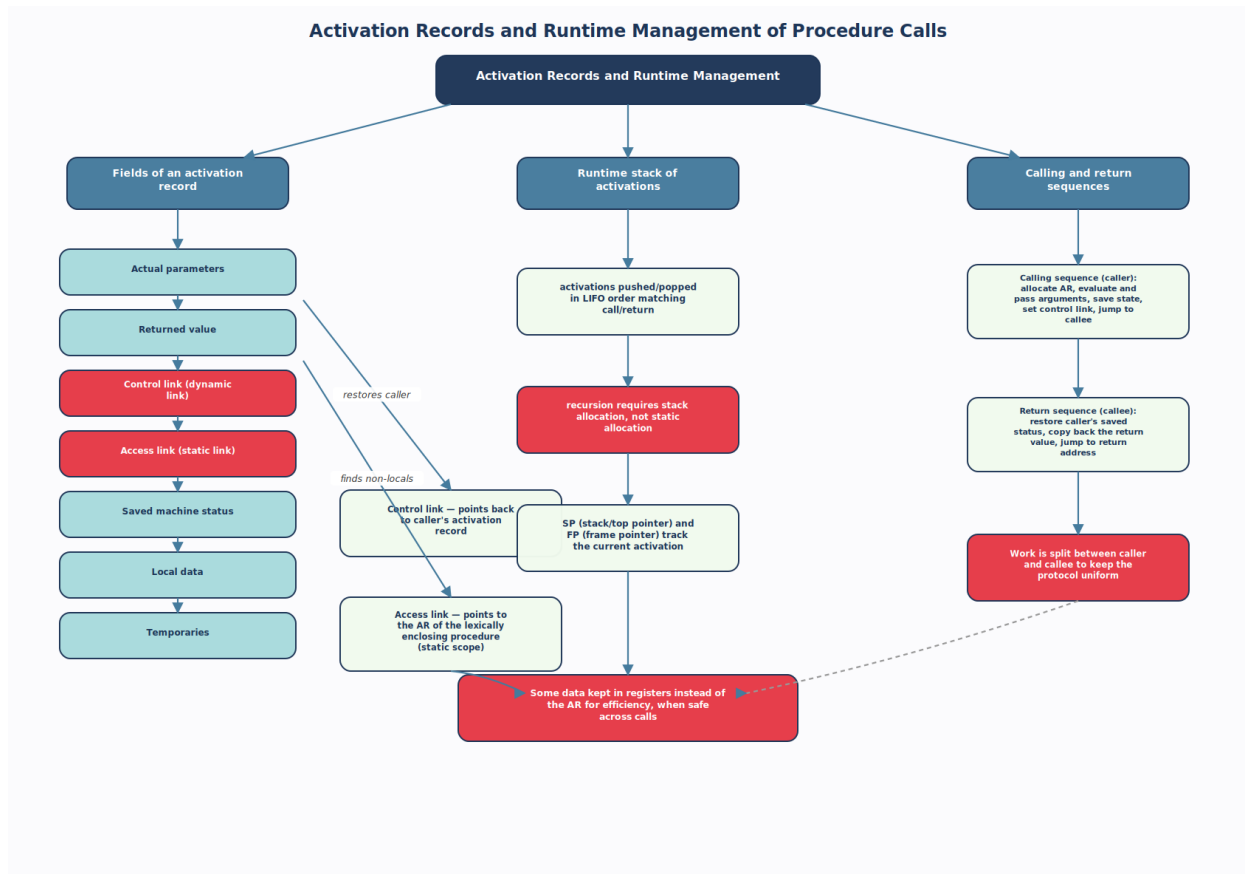
Concept map: param, call, and return three-address statements represent procedure/function invocation, with argument evaluation order and the function-vs-procedure distinction feeding into activation-record layout.

33. Intermediate-Code Representation of Parameter Passing



Concept map: call-by-value, call-by-reference, call-by-value-result, and call-by-name mechanisms and how each is realized through param statements and offset-based access in the callee's activation record.

34. Activation Records and Runtime Management of Procedure Calls



Concept map: the fields of an activation record, the LIFO runtime stack of activations, and the calling/return sequences that cooperate to manage procedure calls at runtime.